

Experiment 1: Student Performance Analysis Using NumPy

AIM

To calculate the average score of each subject using NumPy arrays and identify the subject with the highest average score.

ALGORITHM

1. Import the NumPy library.
2. Create a 4×4 array containing student marks.
3. Arrange the subjects in the order: Math, Science, English, History.
4. Calculate the average of each column using `np.mean()` with `axis=0`.
5. Find the highest average using `np.argmax()`.
6. Display the average score of each subject.
7. Display the subject having the highest average score.

CODE

```
import numpy as np

student_scores = np.array([

    [85, 78, 92, 88],

    [90, 85, 89, 76],

    [75, 80, 85, 90],

    [88, 92, 78, 84]

])

subjects = ["Math", "Science", "English", "History"]

subject_averages = np.mean(student_scores, axis=0)

print("Average score for each subject:")

for subject, avg in zip(subjects, subject_averages):

    print(subject, ":", avg)

highest_index = np.argmax(subject_averages)

print("\nSubject with highest average score:",

      subjects[highest_index])
```

INPUT

Student marks:

Math Science English History

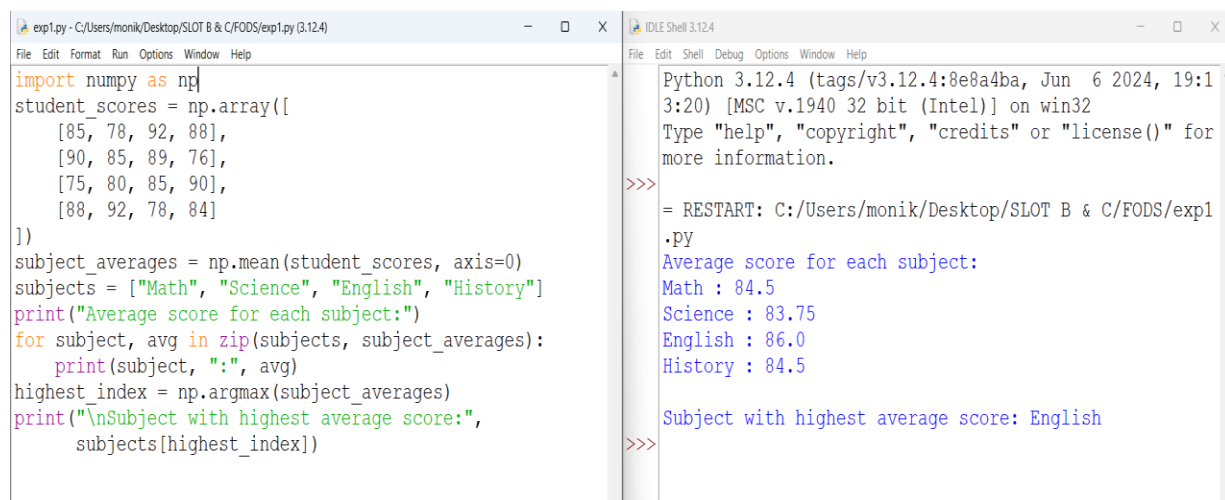
85 78 92 88

90 85 89 76

75 80 85 90

88 92 78 84

OUTPUT

A screenshot of a Python IDE with two windows. The left window, titled 'exp1.py', contains the following Python code:

```
import numpy as np
student_scores = np.array([
    [85, 78, 92, 88],
    [90, 85, 89, 76],
    [75, 80, 85, 90],
    [88, 92, 78, 84]
])
subject_averages = np.mean(student_scores, axis=0)
subjects = ["Math", "Science", "English", "History"]
print("Average score for each subject:")
for subject, avg in zip(subjects, subject_averages):
    print(subject, ":", avg)
highest_index = np.argmax(subject_averages)
print("\nSubject with highest average score:",
      subjects[highest_index])
```

The right window, titled 'IDLE Shell 3.12.4', shows the output of the code:

```
Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun 6 2024, 19:1
3:20) [MSC v.1940 32 bit (Intel)] on win32
Type "help", "copyright", "credits" or "license()" for
more information.
>>>
= RESTART: C:/Users/monik/Desktop/SLOT B & C/FODS/exp1
.py
Average score for each subject:
Math : 84.5
Science : 83.75
English : 86.0
History : 84.5

Subject with highest average score: English
>>>
```

RESULT

Thus, the average score for each subject was calculated successfully using NumPy, and **English** was identified as the subject with the highest average score.